the number of subscribers of current market data aggregators who would become competing consolidators would reduce the consolidation and dissemination fees of those aggregators in equivalent data. For example, some market participants who currently use proprietary data might switch to using consolidated market data products.[2168] Additionally, as discussed below, the availability of the new consolidated market data might allow new entry into the market making, broker-dealer, and other trading businesses.[2169] This would expand the potential subscriber pool, giving competing consolidators a chance to further spread their fixed costs. For these reasons, the Commission believes that the competition among competing consolidators will lead to lower consolidation and dissemination fees for consolidated market data products as compared to these fees for equivalent data today.

c. Connectivity Fees

The Commission believes that connectivity fees charged by competing consolidators for consolidated market data products will also be lower than connectivity fees market participants would currently have to pay to receive equivalent data. To receive data equivalent to consolidated market data today, market participants currently have to pay separately a connectivity fee to the exchanges to access SIP data and a connectivity fee to the exchanges or market data aggregators to access additional data elements that are not part of SIP data but that will be part of consolidated market data. Under the rule, the Commission expects that market participants will pay only one connectivity fee for consolidated market data products (unless they

---

[2168] One of the commenters agreed that some of the current proprietary data users might switch to using consolidated market data products. According to the commenter, a portion of those could become self-aggregators and others could be served by competing consolidators. See Nasdaq Letter IV at 25. See also infra Section V.C.4(a) for a discussion on the effects of the amendments on exchanges' proprietary data feeds.

[2169] See infra Section V.C.4(b) for a discussion on new potential entrants into the market maker, broker-dealer, and other latency sensitive trading businesses.

choose to have a back-up competing consolidator), set by a competing consolidator, and this connectivity fee will be subject to competition among competing consolidators. Competing consolidators will have the ability to sell potentially substitutable data products via their connectivity, subjecting their connectivity products to competition. By contrast, current exchange connectivity fees may not be as competitive because an exchange has sole control over its own connectivity charge for its proprietary market data.[2170] Therefore, the Commission believes that connectivity fees that will be charged by competing consolidators for consolidated market data products will be lower than the connectivity fees for equivalent data today.

The Commission recognizes that SROs will charge connectivity fees to competing consolidators and self-aggregators. The exchanges could continue to set connectivity fees for data feeds as part of their SRO fee schedules, and these fees must continue to meet statutory standards.[2171] The exchanges' connectivity fees are not currently based on the number of end users, and therefore the Commission believes that the connectivity fees for consolidated market data products would not be directly passed through to the end users. SRO connectivity fees would be fixed costs incurred by self-aggregators and by competing consolidators, a cost the latter could spread out among their end users as a part of the consolidation and dissemination as well as connectivity fees.

d. Response to Comments on Fees for Consolidated Market Data

[2170] Several commenters agreed. See, e.g., Virtu Letter at 2; SIFMA Letter at 4; IEX Letter at 5–6.

[2171] For example, under Section 6(b)(4) of the Exchange Act, the rules of an exchange must "provide for the equitable allocation of reasonable dues, fees, and other charges among its members and issuers and other persons using its facilities."

Several commenters stated that the decentralized consolidation model is unlikely to reduce data costs, including because of the richer core data content and the additional upkeep costs introduced by the decentralized consolidation model.[2172] One commenter said that the uncertainty around data reliability and fees did not provide assurances that the market data costs would decline after adoption of these amendments.[2173] Another commenter stated that given the potential data and technology input costs, competition alone cannot lower prices.[2174] Another commenter said that there will be a speed race among competing consolidators and, as a result, as their costs go up their prices will go up.[2175] One other commenter expressed sympathy for the idea of introducing competitive forces, but said that the release did not provide any proof that introducing competition from competing consolidators and self-aggregators will lower data fees and latency.[2176]

The Commission disagrees with these comments and believes that competition among competing consolidators will likely decrease consolidated market data costs for equivalent data.[2177] First, as discussed above, the Commission agrees that there is uncertainty around data content fees. However, for the reasons explained above, the Commission believes that the

---

2172 See, e.g., BlackRock Letter at 5; Cboe Letter at 23–24; Angel Letter at 21; TD Ameritrade Letter at 2; Healthy Markets Letter I at 4; STANY Letter II at 5; Nasdaq Letter III at 8; Nasdaq Letter IV at 8; Data Boiler Letter I at 2.

2173 See TD Ameritrade Letter at 2.

2174 See Healthy Markets Letter I at 4.

2175 See Nasdaq Letter III at 8.

2176 See STANY Letter II at 5.

2177 Several commenters agreed. See, e.g., BestEx Research Letter at 4; Fidelity Letter at 9; Committee on Capital Markets Regulation Letter at 3; Wellington Letter at 1; Intelligent Cross at 5.

overall data fees (i.e., data content, consolidation and dissemination, and connectivity fees) will likely be lower for equivalent data. One commenter's statement about potential fee increases due to "richer core data content"[2178] makes an accurate comparison to baseline SIP data fees difficult, because current SIP data fees bundle data content and consolidation and dissemination fees. While data content portion of the SIP data fees might go up because of the richer content of consolidated market data, consolidation and dissemination portion of SIP data fees could approach zero, as the exclusive SIPs will be discontinued. Thus the overall outcome is unclear, making comparisons to the current SIP data fees very difficult. A more accurate way to examine the data fees is by breaking them down into the three fee components (i.e., data content, consolidation and dissemination, and connectivity) while holding data content constant, as in the Commission's analysis above. As a result of that analysis, the Commission concludes that the overall data fees (i.e., data content, consolidation and dissemination, and connectivity fees) will likely be lower for equivalent data.

Second, the Commission understands that competing consolidators will have input and technology costs, but as discussed above, these are mostly fixed costs that competing consolidators will spread over their customer base. Competitive pressure will encourage competing consolidators to always look for ways to reduce their costs and try to capture market share by passing some or all of these cost savings onto their customers. Additionally, the Commission believes that this same competitive dynamic will be unchanged even if competing consolidators charge different consolidation and dissemination prices for different products, such as higher prices for lower latency products, as suggested by one of the commenters. On the other hand, current market participants whose trading strategies require low-latency data need to buy

[2178] See BlackRock Letter at 5.

proprietary data and the exchanges may not be subject to robust competition in their proprietary data business.[2179] Similarly, the exclusive SIPs are not under competitive pressure and are unlikely to be focused on cost saving measures, as the competing consolidators will.

Some commenters stated that the Commission's belief about multiple competing consolidators offering differentiated products is in conflict with its predictions about the overall fees for consolidated market data potentially being lower than today for equivalent data.[2180] One commenter said that "if the Commission's prediction of 12 consolidators were correct, the fixed costs associated with the two exclusive SIPs would be supplemented with the fixed costs associated with 12 consolidators, likely resulting in a substantial increase in industry fixed costs. Such an increase in fixed costs would ultimately have to be borne by industry participants, including investors, and ultimately recovered from consumers of market data."[2181] Another commenter relied on an academic article to make the point that competition could increase prices when product differentiation is possible.[2182] The Commission does not believe the commenters' conclusions necessarily follow for the new competing consolidator business.

First, the Commission does not believe that the market having several competing consolidators will lead to higher fixed costs for equivalent data and thus higher consolidated market data prices. What the competing consolidators' fixed costs will be is uncertain, because a portion of those fixed costs will be the connectivity fees that the SROs will file with the

---

2179 See supra Section V.B.3(b) for a discussion on the current market structure for proprietary market data.

2180 See Nasdaq Letter IV at 2, 25, 26; Angel Letter at 23.

2181 See Nasdaq Letter IV at 25.

2182 See Angel Letter at 23.

Commission and those are yet to be proposed.[2183] Additionally, even if the fixed costs end up being higher, that would not immediately imply higher consolidated market data prices because the demand for consolidated market data products might be higher as a result of some market participants potentially choosing to buy consolidated market data products instead of proprietary data feeds [2184] and potential new entry into the broker-dealer, market making, and other latency sensitive businesses.[2185] Finally, the Commission believes that, unlike in the current centralized consolidation model, competitive pressures will make it harder for competing consolidators to raise their prices to supra-competitive levels under the decentralized consolidation model. At any given point, competing consolidators are unlikely to have exactly the same incremental costs. Some of them might have cost advantages over the others, which will allow them to pass these cost advantages to customers in terms of lower prices and to compete over each other's customer segments. Those competing consolidators with a cost advantage will increase their market share by pushing out the higher cost competing consolidators from the market. Eventually the market will reach a stable level of competition where individual competing consolidators cannot raise their prices to supra-competitive levels without risking a loss of their customers to a competitor.

2183 See supra Section V.C.2(b)(i)c for a discussion on connectivity fees and their potential impact on competing consolidators.

2184 See infra Section V.C.4(a) for a discussion on the impact of the amendments on proprietary data business.

2185 See infra Section V.C.4(b) for a discussion on the impact of the amendments on new entrants into broker-dealer, market making, and other latency sensitive businesses.

Second, the Commission does not believe that the comment about competition potentially leading to price increases in a market with product differentiation applies to the competing consolidator business. The comment relies on an academic paper that examines prices and competition under a duopoly market structure. The academic paper does not examine or try to understand potential price outcomes under a different market structure, such as one with several competing firms—the most likely outcome for the competing consolidator business.[2186] Therefore, it is hard to extrapolate the paper's arguments from a duopoly market to a market with more than two competitors because the competitive dynamics and the resulting price effects in a duopoly market might be very different from the competitive dynamics and the resulting price effects in a market with several competitors. The Commission acknowledges that competition may not be very strong if the new competing consolidator business ends up with two competing consolidators, especially if both of them are affiliated with the exchanges that currently operate an exclusive SIP.[2187] However, even in such a market, this academic paper's predictions will not necessarily be applicable because the two competing consolidators will have the ability to target each other's customers if they see an economic opportunity to do so. On the other hand, the paper cited by the commenter examines a research question motivated by empirical observations in industries such as the anti-ulcer drug market. Competitors in those markets have a hard time offering each other's products, given potential patent and other restrictions. However, in the consolidated market data business, firms can offer perfectly or partially substitutable products as

---

2186 See Yongmin Chen and Michael H. Riordan, *Price-Increasing Competition*, 39 RAND J. ECON. 1042, 1056 (2008). See also, supra Section V.C.2(a)(ii) for a discussion of the potential scenarios for the number of competing consolidator registrants.

2187 See supra Section V.C.2(a)(ii)b for a discussion on the potential softening of competition if the only two registrants for the new competing consolidator business are exchange-affiliated competing consolidators.

well as each other's differentiated products if an economic opportunity to do so exists. For example, as discussed above, an economic opportunity may exist if an inefficient competing consolidator charges prices above competitive levels. Unlike a brand name drug manufacturer, competing consolidators will maintain the ability to compete over the same customer segments, even as they differentiate their products. And this ability to compete will create a threat of competition that will discipline competing consolidators' prices.

(ii) Fees for the Content of Current SIP data

The Commission also considers the effect of the rule on fees market participants currently pay for SIP data content versus what they would pay for equivalent content under the decentralized consolidation model. The Commission recognizes that a significant proportion of market participants currently purchase only SIP data and/or the unconsolidated equivalent of SIP data.[2188] Under this rule and conditional on fees for consolidated market data, while some of these market participants will choose to purchase more data than they purchase today, other market participants may choose to continue to purchase content equivalent to current SIP data (e.g., NBBO and TOB).[2189] The Commission believes that data fees paid for equivalent data could be similar to current SIP data fees or could be lower than current SIP data fees. Whether the fees are the same or lower depends on several factors: the data content fee structure proposed by the effective national market system plan(s) for NMS stocks, how competing consolidators allocate their costs of processing (i.e., receiving, consolidating, and disseminating) consolidated market data, and any connectivity fees charged by competing consolidators for consolidated market data products.

[2188] Several commenters agreed. See, e.g., BestEx Research Letter at 2–3; State Street Letter at 2; BlackRock Letter at 1.

[2189] Some commenters agreed. See, e.g., MEMX Letter at 5.

a. Data Content Fees

The Commission believes that the data content fee structure proposed by the effective national market system plan(s) for NMS stocks under the decentralized consolidation model is an important factor in determining whether total data fees (i.e., the sum of data content fees, consolidation and dissemination fees, and connectivity fees) for the equivalent of current SIP data could be similar or lower under this rule.[2190] Until the effective national market system plan(s) propose fees for data content underlying consolidated market data offerings, the Commission is unable to determine the extent to which this fee structure would charge lower fees for end users who wish to receive subsets of consolidated market data from competing consolidators.

The Commission also understands that the current SIP data content fees are different for different use cases.[2191] In the 2018 SEC roundtable, several exchanges agreed that there many different types of market participants and that one type of data product does not meet everybody's needs.[2192] The amendments will not change the market reality that market

---

2190 See supra Section V.B.2(c).

2191 See CTA Plan, Q2 2020 CTA Quarterly Revenue Disclosure, available at https://www.ctaplan.com/publicdocs/ctaplan/CTA_Quarterly_Revenue_Disclosure_2Q2020.pdf; Nasdaq UTP Plan, Q2 2020 UTP Quarterly Revenue Disclosure, available at https://www.utpplan.com/DOC/UTP_Revenue_Disclosure_Q22020.pdf.

2192 One of the round table participants said that "there are many different types of market data consumers, from major Wall Street banks and market makers to retail online brokerages and media companies across the world and all have differing data needs." (See Oliver Albers speech on page 107, available at https://www.sec.gov/spotlight/equity-market-structure-roundtables/roundtable-market-data-market-access-102518-transcript.pdf). Another round table participant stated that "[e]xchanges offer a variety of data products to meet the diverse needs of market participants." (See James Brooks' speech on page 177, available at

participants have diverse data needs. Thus the Commission believes that the effective national market systems plan(s) will likely take this market reality into account when proposing the fee schedule for data content underlying consolidated market data by, for example, proposing different fees based on the scope of data content a market participant consumes or usage category or a combination of both.[2193]

Two commenters[2194] emphasized the uncertainty around the potential national market system plan(s) fee schedules, with one commenter stating that "the Commission cannot assume that the operating committee of an NMS plan would create such a [top of book] product, or whether the costs of such a product would meet the needs of market participants who do not want or cannot consume the full consolidated market data."[2195] Indeed the Commission does not assume that the Operating Committee of an effective national market system plan will create a differential pricing structure that might satisfy the needs of market participants who will continue to purchase data content equivalent to the current SIP data.[2196] However, the Commission believes that this is a likely outcome based on market realities. The effective national market system plan(s) may choose different price levels based on usage category because that is the

---

https://www.sec.gov/spotlight/equity-market-structure-roundtables/roundtable-market-data-market-access-102518-transcript.pdf).

2193 One commenter agreed that different types of investor place different values on market data and therefore the market data pricing schemes should take this into account. See Angel Letter at 9, 11, and 27.

2194 See Nasdaq Letter IV at 47; NYSE Letter II at 4.

2195 See NYSE Letter II at 4.

2196 See Proposing Release, 85 FR at 16840–41 for a discussion on the uncertainty about data content fees.